

Internet Path Analysis Using Traceroute

Objective

The purpose of this study is to analyze the path taken by packets from the local computer to a remote destination using traceroute and to examine the intermediate network devices involved in the communication process.

Tools Used

Commands used in this study:

- “tracert”

Step 1 – Route Discovery

The command was executed: tracert google.com

Output:

Tracing route to google.com [142.251.xx.xx] over a maximum of 30 hops:

1 10 ms 7 ms 22 ms 192.168.xx.xx

2 13 ms 17 ms 19 ms 100.64.xx.xx

3 11 ms 29 ms 11 ms 172.25.xx.xx

4 13 ms 14 ms 9 ms 172.25.xx.xx

5 * * 18 ms 172.25.xx.xx

6 * * * Request timed out.

7 23 ms 23 ms 24 ms 46.197.xx.xx

8 24 ms 33 ms 21 ms 74.125.xx.xx

9 26 ms 24 ms 22 ms 172.253.xx.xx

10 28 ms 33 ms 22 ms lcsofb-ab-in-fl4.1e100.net [142.251.xx.xx]

Trace complete.

* Sensitive network information has been anonymized for security purposes.

Findings:

The traceroute results were analyzed to identify the network path between the local computer and the destination host.

- **Total hops observed:** 10
- **First hop:** Local router (Default Gateway)
- **Intermediate hops:** ISP internal routers
- **Final destination:** Google server
- One intermediate router did not respond to traceroute requests (Request Timed Out), which is normal behavior for some network devices.

Analysis:

The traceroute results show that packets first reached the local router before entering the ISP network. Multiple intermediate routers forwarded the packets toward Google's infrastructure. One router did not respond to traceroute probes, resulting in a "Request timed out" message. This behavior is common because some routers are configured to ignore ICMP Time Exceeded messages for security or performance reasons. The packets ultimately reached the destination successfully after traversing ten network hops, demonstrating normal end-to-end routing across the Internet.

Key Concepts Learned

During this exercise, the following concepts were reinforced:

- Router hop concept
- Packet forwarding across multiple routers
- Internet routing path
- ICMP Time Exceeded messages in traceroute
- Network latency measurement
- Difference between local, ISP, and destination networks

Conclusion

The traceroute analysis successfully identified the network path from the local computer to the destination host. The results demonstrated how packets traverse multiple routers across the local network, ISP infrastructure, and the public Internet before reaching the destination. This study also provided practical experience in analyzing network routing behavior and interpreting traceroute results for basic network troubleshooting.